

DisasterMind — Technical & Product Overview

DisasterMind — Technical & Product Overview

An autonomous, multi-agent disaster-coordination platform with validated, real-data risk prediction and a human-in-the-loop command pipeline.

DisasterMind ingests live hazard signals for cyclones/floods, earthquakes, and urban/forest fires; predicts per-asset risk with models validated on real historical events; models cascading effects; plans resources and evacuation routes; and drives a human-commanded decision pipeline that ends in dispatch — all surfaced through a browser command-and-control console.

1. Executive summary

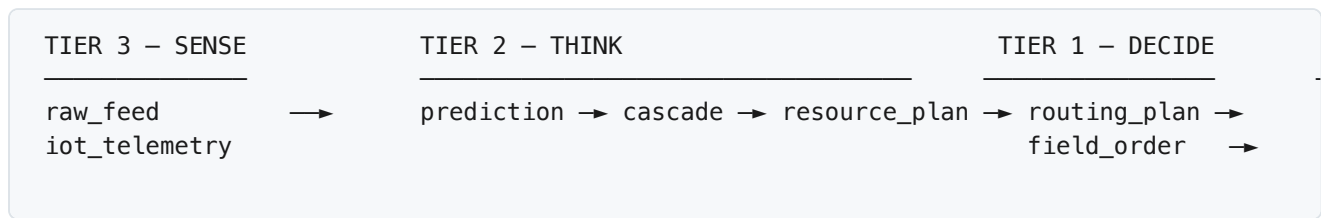
What it is	A decision-support system for multi-hazard disaster early warning and coordination.
Hazards	Cyclone/flood, earthquake (rapid impact assessment), urban/forest fire.
Differentiator	Every model is validated on real historical data , leak-free, and shown to beat the operational baselines with statistical significance — not a demo on synthetic numbers.
Design stance	Standard-library-first and degrades gracefully; explicitly <i>decision-support</i> (a human commander holds authority) with a tamper-evident audit trail.
Scale	~37,500 lines of Python across 34 subsystems, a Vite + React 19 + TypeScript console (67 source modules), and a test suite of ~1,030 offline, deterministic tests.

2. The problem

Disaster response fails at the seams between sensing, prediction, and action: warnings arrive too late to act on, forecasts are accurate on average but blind to the vulnerable, alerts reach phones but not people, and decision-makers are handed a probability with no notion of what to *do* with it. DisasterMind addresses the full chain — **sense** → **predict** → **decide** → **disseminate** — and, critically, treats *validation and honesty* as first-class engineering, so its outputs can earn the trust an evacuation decision requires.

3. System architecture

DisasterMind is a **three-tier multi-agent system** communicating over a single message bus. A hazard signal flows up the tiers and back down to dispatch:



- **Tier 3 (sensing & acting):** ingestion adapters for each hazard, IoT gateways, social-signal intake, and outbound dispatch.
- **Tier 2 (analysis & planning):** per-hazard prediction agents, cascading-effect models, resource optimisation, evacuation routing, and field tasking.
- **Tier 1 (command):** the commander agent, escalation logic, and concurrent multi-incident orchestration.

Message bus & topics. All agents publish/subscribe over an in-memory `MessageBus` on typed topics (`raw_feed`, `iot_telemetry`, `prediction`, `cascade`, `resource_plan`, `routing_plan`, `field_order`, `commander_review`, `escalation`, `dispatch`), with five priority levels (CRITICAL→INFO). A coordination loop ticks on a fixed cadence, builds the agent DAG, and routes every message through all stages.

Hazard modules. Three domains run as independent pipelines: **A — cyclone/ flood**, **B — earthquake**, **C — fire/structural collapse** (plus cross-module coordination).

Graceful degradation. The core runtime is standard-library only. Heavy capabilities (gradient-boosted models, OR-Tools, Postgres/Timescale, Kafka, satellite feeds) are optional; when absent, the system falls back to deterministic in-process implementations rather than failing.

4. Capabilities in detail

4.1 Multi-hazard risk prediction

Per-hazard models score per-asset/per-cell risk from physically meaningful drivers, each with explainable feature attributions (SHAP-style): - **Earthquake** — magnitude, depth, location physics → measured damage-grade outcome (instrumental intensity / loss alert). - **Cyclone/flood** — rainfall accumulations, river discharge and trend, surge, seasonality → flood-threshold exceedance. - **Fire** — fire-weather (temperature, humidity, wind), drought/dryness streaks, seasonality → next-day ignition.

Models are thin wrappers that prefer a trained gradient-boosted/logistic backend and fall back to a deterministic in-process model, always returning calibrated probabilities.

4.2 Cascading-effect modeling

Beyond first-order risk, the platform models downstream consequences: earthquake **aftershock sequences** (Omori-law decay), **flood propagation** along river networks, and fire spread — so the coordinator anticipates the second wave, not just the initial event.

4.3 Resource allocation & evacuation routing

- **Resource optimiser** allocates scarce assets (teams, vehicles, supplies) across competing demands.
- **Road-network routing** plans evacuation routes over real OpenStreetMap road graphs to real shelter locations, with road-closure awareness.

4.4 Evacuation decision layer

The component that turns a forecast into an actionable order: - **Clearance-time model** — given zone population, road egress capacity, and shelters, computes how long the zone takes to empty and *by when the order must be issued*; flags zones that cannot be cleared in the available lead time and recommends vertical (in-place) evacuation; supports contraflow and egress sensitivity analysis. - **Vulnerability-weighted phased evacuation** — schedules cohorts (hospitalised, elderly/mobility-impaired, transport-dependent, general public) against scarce assisted-transport capacity and surfaces exactly which group is left behind when warning time runs short. - **Dissemination & warning-response modeling** — multi-channel reach (cell broadcast, SMS, siren, radio, community wardens) combined with a *compliance* model that consumes both lead time and false-alarm rate (capturing the cry-wolf effect), reporting who is left behind and *why* (not reached vs. won't comply vs. cannot be moved). - **Cost/benefit & break-even** — net lives saved, evacuation-induced casualties, and the probability threshold below which ordering an evacuation costs more than it saves (over-evacuation guard). - **Integration capstone** — one per-zone recommendation (`ORDER_BY_DEADLINE` / `NOT_CLEARABLE_VERTICAL` / `BELOW_BREAKEVEN_HOLD` / `NO_ACTIONABLE_WARNING`) plus a defensible decision record (“what we knew, when, and what we recommended”).

4.5 Human-in-the-loop command & escalation

The commander tier reviews agent recommendations, manages an escalation queue with approve/reject/timeout semantics, generates escalation memos and situation briefings, and runs concurrent incidents (one coordination DAG per active disaster). The system never issues an autonomous evacuation order; it recommends, and a human acts.

4.6 Field operations

Device-facing field contracts and a client that close the dispatch → acknowledge → GPS-update loop, with a durable outbox (terrestrial first, satellite fallback, offline queue) for connectivity-denied environments.

4.7 Emergency alerting

Standards-compliant **CAP 1.2** emergency-broadcast XML generation for interoperability with public warning systems.

4.8 Validation & evidence framework (*the platform's signature capability*)

A complete, dependency-free evaluation suite that scores and documents every model on real, held-out data: - **Metrics** — rank-based ROC-AUC, Brier score, accuracy, calibration/ECE. - **Decision-point metrics** — POD (detection), FAR (false-alarm ratio), CSI, frequency bias, HSS at a chosen operating threshold, with explicit miss-vs-false-alarm cost accounting. - **Operational-baseline comparison with significance** — paired-bootstrap tests (p-values, confidence intervals) against the incumbents a forecaster uses today. - **Lead-time-vs-POD curves** — the *actionable warning time*, not just accuracy. - **Blocked cross-validation** — leave-one-region-out and rolling-origin, reporting the worst block, not just the average. - **Calibrated uncertainty** — isotonic recalibration and split-conformal prediction sets with verified coverage. - **Fairness audit + remediation** — per-subgroup detection rates, with equalized-odds remediation that states the false-alarm *cost of equity* and classifies residual gaps by cause. - **Rare-severe-event (tail) analysis** with bootstrap confidence intervals. - **Drift detection & retraining triggers** — PSI/KS feature drift and a skill-decay-driven retrain decision. - **Degraded-input robustness** — graceful-degradation curves under sensor failure. - **Shadow mode** — an append-only, hash-chained journal that records live predictions before outcomes are known, scores a season against reality, and exports a complete record for independent review.

4.9 Historical hindcasting & backtesting

- **Named-event replay** — Cyclone **Fani (2019)** and **Amphan (2020)** are replayed from their real best-tracks using only pre-cutoff data (leak-free), driving the full activation and coordination pipeline and scoring against the documented outcome (people evacuated, fatalities, landfall intensity).
- **Population-scale cyclone backtest** — the entire modern North-Indian-Ocean record: **92 named, India-landfalling cyclones (1990–2025)** from NOAA IBTrACS, scored for activation lead time and landfall-extrapolation error.
- **Full-pipeline backtest** — forecast → evacuation decision → scored against reality, on real storms.

4.10 Live data ingestion

Adapters target authoritative live sources: **USGS** and **India NCS** seismology, **Open-Meteo** (GloFAS flood + weather), **NASA FIRMS** active-fire detections, **India-WRIS** river monitoring, **ISRO Bhuvan** flood inundation, and **IMD** forecasts. Several operate live and key-free today; the remainder are wired and require provider keys/endpoint configuration.

4.11 Production engineering

Health/readiness probes, retry and circuit-breaker resilience, graceful shutdown; API authentication, rate limiting, payload validation, CORS hardening; metrics collection with Prometheus exposition; distributed tracing with per-incident latency; a tamper-evident audit chain with signing, retention, and backup; durable storage (Postgres/Timescale, Elasticsearch, MinIO) with versioned migrations; cross-district mutual-aid federation; a structured-logging stack; and a system self-check (“doctor”) plus a load/throughput benchmark harness.

4.12 Web command console (clients/web)

A unified Vite + React 19 + TypeScript application (MapLibre GL maps, Recharts) with four modules — **Commander Dashboard**, **Escalation**, **Field Ops**, and **Post-Incident Report** — talking to the platform API (configurable base URL), with live status, override controls, SHAP explanations, and PDF report export.

5. Validation results (real data, leak-free, out-of-sample)

All figures are produced by the validation suite on committed real-data fixtures with strictly temporal splits; operating thresholds and calibrators are fit on a calibration split, never on the test set.

Hazard	Data source	Out-of-sample AUC	Brier	ECE	Actionable lead (POD ≥ 80%)
Earthquake	USGS catalog (2013–2017, M4.5+)	0.937	0.011	0.002	n/a (instantaneous)
Flood	GloFAS-ERA5, 12 Indian basins (2010–2023)	0.944	0.028	0.004	168 h (7 days)
Fire (PNW)	USDA FPA-FOD + ERA5 (2012–2018)	0.837	0.121	0.023	72 h (3 days)
Fire (India)	NASA FIRMS VIIRS + ERA5 (2019)	0.80 (raw)	0.166	0.043	seasonal*

Beats the operational incumbents, with statistical significance: - **Flood** beats *persistence* (the standard no-model hydrological forecast, $p \approx 0.004$) and *seasonal climatology* ($p \approx 0.004$). - **Fire** beats the *Angström fire-danger index* — both on US ($p \approx 0.004$) and on real Indian data ($p \approx 0.02$). - **Earthquake** statistically matches a GMPE ground-motion attenuation baseline on the damage label and **beats USGS PAGER by +0.22 AUC** on the felt-report label.

Population-scale cyclone evidence (92 real storms, IBTrACS): the system would have raised a cyclone alert a **median of 54 hours before landfall**, with **≥48 h lead for ~58%** of storms and **≥72 h for ~40%**.

Operational decision quality is reported per hazard at the dispatch threshold (POD/FAR/CSI), with calibration repaired by isotonic recalibration (e.g., earthquake ECE 0.21 → 0.002), conformal coverage at target, blocked cross-validation worst-block scores, a published fairness audit with remediation, and rare-severe tail analysis.

* The India fire model is validated on a single FIRMS year via a leak-free intra-year split; its strong seasonal base-rate shift makes the fixed-threshold operating point conservative while discrimination (AUC) remains solid.

6. Data sources (committed, reproducible)

Domain	Source	Coverage
Earthquakes	USGS FDSN catalog	~36,500 real M4.5+ events, 2013–2017
Floods	GloFAS-ERA5 discharge + ERA5 rainfall (Open-Meteo)	12 Indian river-basin sites, 2010–2023
Fire (US)	USDA FPA-FOD occurrences + ERA5	12 Pacific-Northwest cells, 2012–2018
Fire (India)	NASA FIRMS VIIRS detections + ERA5	10 Indian fire-belt cells, 2019
Cyclones	NOAA IBTrACS best tracks	92 named India-landfalling storms, 1990–2025
Cyclone cases	IBTrACS + documented outcomes (IMD/EM-DAT)	Fani 2019, Amphan 2020
Infrastructure	OpenStreetMap road graph, Copernicus DEM, Census	Puri / Fani impact zone
External outcomes	GDACS (UN/EC) declared disasters	166 Indian flood/cyclone events

All data is real and citable; every fixture records its provenance. The validation suite runs fully offline against committed fixtures; a separate fetch utility rebuilds them from the free public APIs.

7. Technology & deployment

- **Language/runtime:** Python ≥ 3.11 , standard-library-first; optional extras for ML (XGBoost/scikit-learn/SHAP), optimisation (OR-Tools/PuLP), geospatial, messaging (Kafka), storage (Postgres/Elasticsearch/MinIO), and feeds.
- **API:** framework-free dashboard service over HTTP + WebSocket (health/ readiness probes, topics, incidents, escalations with approve/reject, live stream), OpenAPI-specified.
- **Frontend:** Vite + React 19 + TypeScript, deployable to Vercel.
- **Packaging/CI:** multi-stage Docker image (pinned base, non-root, healthcheck); GitHub Actions pipeline with a Python test matrix, lint, type-check, security scanning (Trivy + pip-audit), and a CycloneDX SBOM.
- **Deployment targets:** container/Railway and Kubernetes manifests with a SQL schema.
- **CLI:** `run`, `simulate`, `train`, `eval`, `doctor`, `serve`, `verify-audit`, plus dedicated entry points for validation, hindcasting, diagnostics, and the demo.

8. Maturity & roadmap

Current status — a validated research-grade platform. The prediction and evaluation layers are rigorous and reproducible; the coordination and evacuation- decision layers are complete and operate end-to-end on real historical events.

Known limitations (stated transparently): - Earthquakes cannot be *forecast* on an evacuation horizon; the earthquake module performs rapid impact assessment, and the evacuation framing applies to cyclone/flood/fire. - Evacuation-planning parameters (clearance times, compliance rates, casualty rates) are explicit, tunable planning assumptions pending calibration against agency ground truth. - Some outcome labels are well-justified proxies (discharge exceedance, instrumental intensity, satellite detections) rather than surveyed losses. - The India fire model is currently single-year; multi-year expansion is data-availability-bound.

Roadmap to operational deployment: 1. Calibrate the evacuation layer against district-level historical response data. 2. Complete live-feed integration (provider keys + endpoint configuration) and expand the India fire record to multiple years. 3. Run a full **shadow-mode season** in partnership with a state disaster- management authority — predicting live, acting on nothing, scored against actual outcomes — followed by independent review. This is the decisive step toward trusted operational use.

DisasterMind — earlier, better-targeted, and more equitable disaster warnings, built to earn the trust an evacuation decision requires.